

Rudra Patel

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Education

Lewis University

May 2026

B.S. Computer Science, Concentration in Artificial Intelligence - GPA 3.9, Dean's List 6x

Coursework: Data Structures & Algorithms, Computer Organization (C, C++), Big Data Systems (Apache Kafka, Spark)

Experience

Research Assistant

April 2026 - June 2026

Lewis University

- Built an internal dataset management platform with Go, React, and PostgreSQL to centralize dataset upload, documentation, search, editing, and download workflows for Lewis Research Lab
- Implemented large-file upload and download workflows in Go using concurrent chunked uploads and streaming, supporting files up to 50GB without loading them fully into server memory
- Developed dataset search and filtering workflows across keywords, labels, summaries, and file metadata, helping researchers find and reuse relevant datasets more efficiently
- Designed an automated metadata pipeline that profiles uploaded datasets and generates labels, class distributions, summaries, and suggested search terms to make research data easier to understand and reuse

Software Engineer

May 2025 - May 2026

Hometown Liquor - E Commerce Platform

- Built a production e-commerce platform using React, Java Spring Boot, and PostgreSQL, supporting 150+ users, 770+ products, product search, and inventory workflows while helping grow monthly revenue to \$25K
- Secured 92+ REST API endpoints with Spring Security, stateless authentication, role-based admin access, request limits, and activity logs to protect live customer and store data
- Automated daily inventory sync with a third-party POS vendor API, processing products in 100-item batches, handling duplicate records, updating stock quantities, and protecting reliable store data from incorrect overwrites
- Improved release reliability with Docker containers, CI/CD, and 85+ unit and integration tests covering authentication, products, inventory, and core API workflows

Projects

LocalCI - Cross-Platform CI Runner | Go, Python, GitHub Actions, YAML

[Link](#)

- Built an open-source CI runner in Go that lets developers reproduce GitHub Actions-style workflows locally by validating YAML pipelines and running dependent jobs from the terminal on Windows, macOS, and Linux
- Implemented parallel job execution with dependency handling, allowing independent jobs like linting, unit tests, and integration tests at the same time, cutting benchmark runtime by 47.4%
- Improved failure debugging by saving logs and artifacts for every job, generating HTML reports, rerunning failed jobs only, and grouping repeated failures to identify persistent problems

Real-Time Event Processor | Node.js, Express.js, TypeScript, Redis, TimescaleDB, Docker

[Link](#)

- Built a real-time analytics pipeline similar to Kafka with Redis Streams, Node.js, TimescaleDB, and React to ingest, process, store, and visualize 2.4M+ events in 24 hours with zero observed message loss
- Implemented a Redis Stream batch consumer that grouped up to 100 events per transactional write, reducing database transactions by 60-70% while preventing partial data storage during failures
- Replaced dashboard scans over millions of event rows with precomputed TimescaleDB aggregates, reducing analytics query latency by 55x from 1.8s to 33ms

Recipe Backend Service | Java, Spring Boot, AWS, PostgreSQL, Docker, Terraform

[Link](#)

- Built a Java Spring Boot backend with 12 REST API endpoints for recipe generation, history, and admin workflows, secured with stateless JWT auth, BCrypt password hashing, and per-user access control
- Defined AWS infrastructure in Terraform with ECS Fargate, RDS, IAM, and CloudWatch, then validated reliability with 139 unit and integration tests covering auth, validation, failure handling, and S3 exports

Technical Skills

Languages: Java, TypeScript, Go, Python, C++

Backend & Frontend: Spring Boot, React, Node.js, Express.js, REST APIs

Data & Infrastructure: PostgreSQL, MySQL, Apache Kafka, Apache Spark, Redis, TimescaleDB

Cloud & DevOps: AWS ECS, AWS RDS, AWS S3, AWS CloudWatch, AWS IAM, Terraform, Docker, Kubernetes